



Refractive index of soft contact lens materials measured in packaging solution and standard phosphate buffered saline and the effect on back vertex power calculation

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ABSTRACT

Purpose: To measure the refractive index (RI) of commonly available soft contact lens (CL) materials, their packaging solutions and compare to the manufacturers' nominal RI. The relationship between RI versus water content, and the effect of inaccurate RI when converting lens power measured in solution to in-air back vertex power were examined.

Methods: The RI of 18 single vision soft CL materials were measured using CLR 12–70 digital refractometer. Three lenses of each material were measured, in their packaging solution and then after soaking in standard phosphate buffered saline (PBS). The RIs of packaging solution were also measured. Accuracy requirements for correct wet to dry power conversion based on thick lens formula were projected.

Results: The standard deviation between three samples was less than 0.005. The measured RI ranged from 1.3744 ± 0.001 – 1.4265 ± 0.0004 for PBS soaked and from 1.3739 ± 0.0003 – 1.4264 ± 0.0024 for packaging solution soaked materials. Comparing nominal with mean measured PBS and packaging solution RIs, 5 and 3 lens materials, respectively, fell outside ISO tolerance. The packaging solution RI of DailiesAquaComfortPlus had the largest difference of 0.0040, compared to RI of standard PBS. For converting lens power measured in PBS to in-air power, the difference between measured and nominal RI of 0.0104 would result in wrongly calculated in-air power 0.99 D for a -6.00 D lens.

Conclusion: The CLR 12–70 is reliable and accurate refractometer for the measurement of soft CL materials. Accurate RI measurements are of relevance with increased use of wavefront sensors to measure lens power while they are immersed in solution. Even small errors in solution or material RI can lead to significant errors in converted in-air power. To obtain valid in-air lens power results, measurement conditions must match the material and solution RIs used for the conversion.

1. Introduction

Refractometry is the method used to measure the refractive index (RI) of a substance, whereby the RI is defined as the ratio of speed of light in vacuum to the speed of light in the substance. RI is a unitless number, generally dependent on the temperature and also the wavelength. The change in RI due to a change in wavelength is called chromatic dispersion. The RI of a contact lens material is an important physical parameter that affects the lens design, as the refracting power of a lens for any given thickness and curvature is directly related to its RI. It is also an indirect measure of the water content of a soft contact lens, as for most of the hydrogel lens materials, the RIs are closely correlated with the water content. The standard method of measuring

the RI of contact lens materials uses a calibrated Abbe refractometer either at wavelength 546 nm or 587.6 nm as prescribed in ISO 18369-4 [1]. The Abbe refractometer is a widely used laboratory benchtop instrument, which is based on measuring the critical angle. According to the critical angle principle, it measures the angle of the incident light beam at which the beam switches from refraction to total internal reflection at the prism/sample interface. Beside the common critical angle principle used in refractometers, interferometric refractometers are also available where interferometry is used to measure the speed of light in different media. There are two previous studies reported where Linnik micro-interferometer and low coherence interferometer was used to measure RI. For Linnik micro-interferometer, a white light was used to generate a series of colored fringes to measure RI [2]. For low

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coherence interferometer, it is based on measuring the change in the optical distances between two fixed interfaces when material of interest is inserted into the gap between these interfaces.

The Abbe refractometers are still the most commonly used refractometers in which both digital and handheld instruments are available. The handheld refractometers are generally less accurate and are not equipped to control the temperature. These instruments are unsuitable to determine dispersion as they use ambient white light as their illumination source. Typically, a handheld digital refractometer can read to a precision of 0.001 units, compared to a resolution of up to 0.0002 for a bench Abbe refractometer. Under experimental conditions, in-vivo RI measurements using a handheld refractometer mounted on a custom slit lamp showed comparable results to in vitro measurements. [4] This has been done using one hydrogel and one silicone hydrogel contact lens to determine whether the RI, and thereby the water contents, of a soft contact lens can be evaluated using refractometry.

The CLR12-70 (Index Instruments Ltd., Cambridge, UK) is an automated digital refractometer designed to measure the RI of soft and rigid contact lens materials. This instrument provides direct readings and is simple to use with minimal influence by the operator. A few previous studies reported the RI results using this instrument showing excellent reliability and validity of RI measurements for hydrogel lens materials [5] and silicone hydrogel lens materials [6,7]. The effect of lens wear on RI by measuring before and after wearing of hydrogel and silicone-hydrogel contact lenses was also assessed using CLR12-70 refractometer [8].

The accuracy of RI measurements are of critical importance when lens power is measured with lenses immersed in solution and conversion to in-air power is required. The ISO 18369-4 [1] states that the back vertex power of hydrogel lens types can be measured in-air using a focimeter or in solution using a Moiré deflectometer or a Hartmann method, with the advantages and disadvantages reviewed previously by Pearson [9]. For the lenses measured in solution, accurate information of the RI of the lens material and the immersion solution is essential for in-air back vertex power conversion [9]. The manufacturer reported nominal RI is usually only in two decimal places, which is not sufficient for accurate power conversion, particularly for higher powered lenses.

The International Standard Organisation document also states that standard phosphate buffered saline (PBS) [10] solution should be used for contact lens testing with pH of 7.4 ± 0.1 and nominal osmolarity of 310 ± 5 mOsm/kg. However, in the subclause it states for technical reasons, some manufacturers pack their lenses in a packaging solution that has a different composition from the 'standard' saline solution. Therefore, to measure the 'label' power, measurements will need to be made using packaging solution instead of the 'standard' PBS solution. As the packaging solution can affect the material refractive index, it is important that the correct refractive indices of material and solution are known and used for the power conversion.

The purpose of this study was to measure RI of all commonly available soft contact lens materials soaked in packaging solution and PBS using the digital CLR 12-70 refractometer at 546 nm wavelength. A comprehensive and up to date table with comparable results of PBS and packaging solution soaked RIs are presented. The RIs of the packaging solution for all lens types were measured and compared to the standard PBS solution. The effect of using inaccurate RIs for the lens and/or the solution when converting lens power measured in solution to the in-air power, based on the thick lens formula, are examined and quantified. The relationship between RI versus water content (WC) for hydrogel and silicone hydrogel lens materials is also examined.

2. Materials and methods

2.1. Instruments and analysis

The automated digital CLR 12-70 refractometer was used to measure the RI of soft contact lens materials. This instrument measures RI

using back reflection. The measurement wavelength had been modified by the manufacturer Index Instruments Ltd to 546 nm instead of the standard 589 nm. While the ISO standard allows the use of either wavelength for contact lenses, for intra ocular lenses, all measurements need to be performed at 546 nm. Hence most manufacturers of power mapping instruments use this wavelength. To eliminate possible errors when converting measured to in-air power due to dispersion effects, the RI measurement wavelength was matched to the wavelength used by the power mapping instruments. The CLR 12-70 has a stated and verified accuracy and reproducibility of $RI \pm 0.001$ and $RI \pm 0.0003$ respectively for soft contact lens materials. There are four different operating modes – continuous, stability, autoprint and single scan. The 'single scan' mode was used and the reading of the RI was taken after stabilisation was reached for each lens. Other instrument settings include: emulsion type sample with stability default time set to 60 s.

While the traditional lens power measurements are performed with the lens in air using a focimeter instrument, most of the state of the art power mapping instruments utilize either the Phase-Shifting Schlieren, Hartmann-Shack or Moiré Fringe principle to measure lens power with the lens immersed in solution. The measured power is subsequently converted to in-air power for which accurate knowledge of material and solution refractive indices is required. The discrepancy of using inaccurate RIs when converting in-solution measured lens power to the in-air power, was assessed using the following equation [11,12]:

$$BVP_{air} = \frac{n_{lens} - n_{air}}{n_{lens} - n_{saline}} * BVP_{solution}$$

where n_{lens} is refractive index of the lens material, n_{air} is refractive index of air = 1, n_{saline} is refractive index of the solution the lens is soaked in, BVP_{air} is back vertex power in air and $BVP_{solution}$ is back vertex power measured in solution.

As a first step, the nominal in-air lens powers of -1.00 D, -3.00 D and -6.00 D lenses were converted to in-solution power using PBS measured RIs of lens materials in PBS solution. This calculated lens power in solution was then converted back to in-air power using RI specified by the manufacturer for the lens material and using RI of PBS solution. The same procedure was performed using measured RIs of lens materials in packaging solution. The difference between the nominal in-air power and the re-calculated in-air power using RI measured in PBS and RI measured in packaging solution is presented as the error in the back vertex power.

2.2. Contact lenses

Eighteen types of commercially available single vision soft contact lenses, 7 hydrogel and 11 silicone hydrogel materials, from four leading manufacturers were measured in prescription power of -1.00 D (see Table 1).

2.3. Measurements

All lenses were first measured straight out from the blister packs in which they were soaked in their packaging solution. Lenses were then soaked in PBS for at least 24 h before repeating the measurements. A single RI measurement was performed on three lenses from each lens type at room temperature of $20^\circ\text{C} (\pm 1^\circ\text{C})$. The RIs of the packaging solution and the PBS were also measured for all types. All measurements were performed by a single operator.

2.4. Statistical analysis

A paired *t*-test (two-tailed) was used to compare between RI measured in packaging solution and PBS. The same test was also used for comparing the RIs of packaging solution to the standard PBS. The relationship between RI and WC was assessed using the Pearson's correlation coefficient (*r*). A *p*-value < 0.05 was considered statistically significant. All analyses were performed using Graph Pad Prism 5.

Table 1

List of commercially available soft single vision contact lenses used in this study.

Brand	Manufacturer	Material/RI	BC	Dia (mm)	Water content
Biotrue ONEday	Bausch & Lomb, USA	(Hydrogel) nesofilcon A / 1.374	8.6	14.2	78 %
Dailies Aqua Comfort Plus	ALCON, USA	(Hydrogel) nelfilcon A / 1.38	8.7	14.0	69 %
Proclear	Cooper Vision, USA	(Hydrogel) omafilcon B / 1.39	8.6	14.4	62 %
Proclear 1-Day	Cooper Vision, USA	(Hydrogel) omafilcon A / 1.387	8.7	14.2	62 %
SofLens Daily	Bausch & Lomb, USA	(Hydrogel) hilafilcon B / 1.4036	8.6	14.2	59 %
Acuvue 2	Vistakon, Johnson & Johnson, USA	(Hydrogel) etafilcon A / 1.40	8.3	14.0	58 %
Acuvue Moist 1-Day	Vistakon, Johnson & Johnson, USA	(Hydrogel) etafilcon A / 1.40	8.5	14.3	58 %
Clariti 1-Day	Cooper Vision, USA	(Silicone Hydrogel) somofilcon A / 1.4003	8.6	14.1	56 %
Acuvue TrueEye 1-Day	Vistakon, Johnson & Johnson, USA	(Silicone Hydrogel) naraafilcon A / 1.41	8.5	14.2	54 %
MyDay	Cooper Vision, USA	(Silicone Hydrogel) stenfilcon A/1.40	8.4	14.2	54 %
Biofinity	Cooper Vision, USA	(Silicone Hydrogel) comfilcon A / 1.40	8.6	14.0	48 %
Avaira	Cooper Vision, USA	(Silicone Hydrogel) enfilcon A / 1.40	8.5	14.2	46 %
PremiO	Menicon	(Silicon Hydrogel) asmofilcon A / 1.423	8.6	14.0	40 %
Acuvue Oasys	Vistakon, Johnson & Johnson, USA	(Silicone Hydrogel) senofilcon A / 1.42	8.8	14.3	38 %
PureVision2	Bausch & Lomb, USA	(Silicone Hydrogel) balafilcon A / 1.426	8.6	14.0	36 %
Dailies Total 1	ALCON, USA	(Silicone Hydrogel) delefilcon A / 1.42	8.5	14.1	Gradient 33 % to ≥80 %
Air Optix Aqua	ALCON, USA	(Silicone Hydrogel) lotrafilcon B / 1.42	8.6	14.2	33 %
Air Optix Aqua Night & Day	ALCON, USA	(Silicone Hydrogel) lotrafilcon A / 1.43	8.6	13.8	24 %

3. Results

The standard deviation between three samples was generally less than 0.005 which is the ISO 18369-2 [13] tolerance limit. The measured RIs ranged from 1.3744 ± 0.001 – 1.4265 ± 0.0004 for the PBS soaked measurements and ranged from 1.3739 ± 0.0003 – 1.4264 ± 0.0024 for the packaging solution soaked measurements (see Fig. 1). Comparing the nominal RIs with the means of PBS and packaging solution soaked measured RIs, 5 and 3 lens materials respectively fell outside the ISO tolerance, with the largest difference of 0.0104 for the Proclear 1-Day lens material. Comparing the measured RIs between PBS soaked and packaging solution soaked measurements, the means for seven lens types were significantly different.

The packaging solution RI for Dailies Aqua Comfort Plus lens had the largest difference of 0.0040 compared to the standard PBS RI of 1.334. The packaging solution RIs of all the lens types were higher than the PBS RI of 1.334 (see Fig. 2).

Based on the equation explained in the method section, the in-air power differences between PBS soaked measured RI and manufacturer specified RI are shown in Fig. 3a. The differences of the in-air power calculated for packaging solution soaked measurements between measured RI and specified RI from manufacturer are shown in Fig. 3b. For example, when converting the lens power in PBS to the in-air power, the difference of 0.0104, between using measured and labelled

nominal RI for the Proclear 1-Day lens would result in wrongly calculated in-air power of 0.49 D for a -3.00 D lens and 0.99 D for a -6.00 D lens.

The relationship between RI and WC for hydrogels and silicone hydrogels are shown in Fig. 4. In general, as the WC increased, the RI tended to decrease. For measured RI soaked in PBS, the Pearson's correlation coefficient (r) was -0.932 and statistically significant with p -value < 0.000 . For measured RI soaked in packaging solution, the Pearson's correlation coefficient (r) was -0.966 and, statistically significant with p -value < 0.000 (see Fig. 4a). Similar correlation existed for hydrogels and silicone hydrogels analysed separately (see Fig. 4b).

RI of four silicone hydrogel lens types (Acuvue TrueEye 1-Day, PremiO, Acuvue Oasys and Dailies Total 1) could not be measured, neither soaked in PBS nor in packaging solution, using the CLR12-70 refractometer and hence were omitted.

4. Discussion

In this study, RIs of commonly prescribed soft contact lens materials were measured using the digital CLR 12-70 refractometer at wavelength of 546 nm. The CLR 12-70 refractometer is the only automated refractometer currently on the market and its accuracy is relatively unaffected by operators' skills. Conventional Abbe-type refractometers,

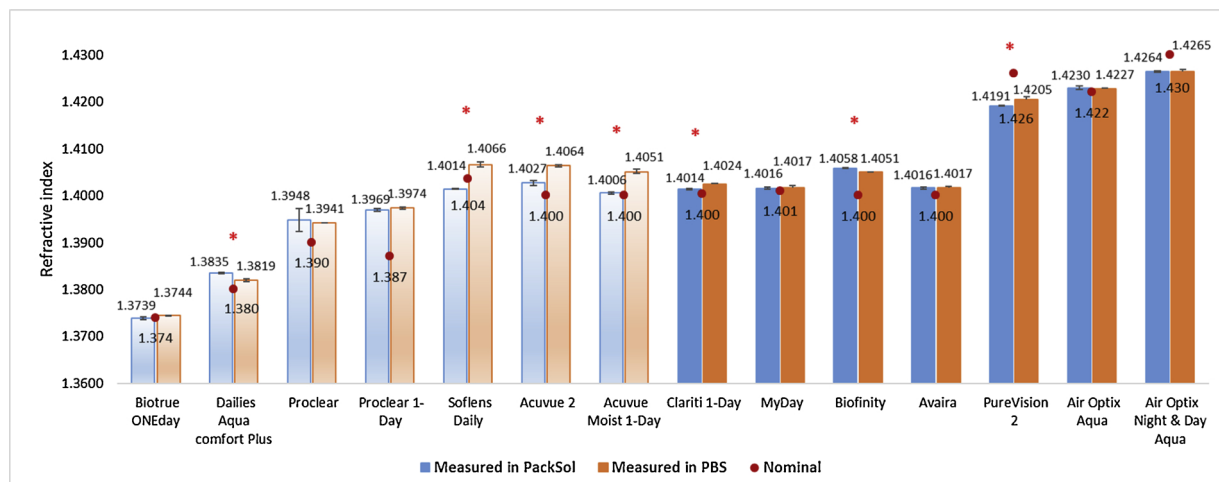


Fig. 1. RI in PBS and packaging solution soaked measured values. The error bars represent standard deviations between three lenses of each lens types. Statistically significant difference between measured RI soaked in PBS and soaked in packaging solution (*).

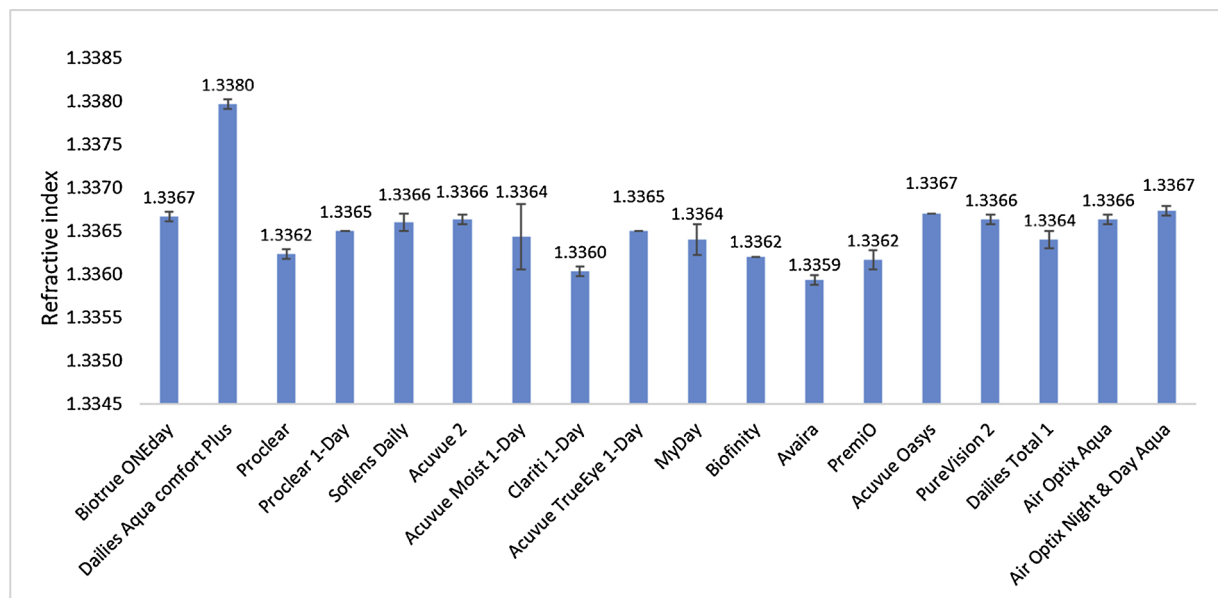


Fig. 2. RIs of the packaging solution for each lens types. The error bars represent standard deviations between measurements of three blister packs.

such as the Atago CL-1, Atago N-2E, Atago N3000 or Metricon are still widely used to measure the RIs of contact lens materials, but they require experienced operators to produce reliable results. Nichols et al. [5] investigated the CLR 12-70 refractometer and found excellent reliability within and between operators for measuring RIs of soft contact lens materials.

The absolute accuracy of the instrument claimed by the manufacturer is of ± 0.0001 RI for typical samples. In this study, the standard deviation between three samples measured for each lens type was generally less than 0.005 which is the ISO 18369-2 tolerance limit. Generally, the CLR 12-70 refractometer measures reliable and accurate data, however it failed to measure and give a reading for four of the silicone hydrogel lens materials - Acuvue TrueEye 1-Day (narafilecon A), PremiO (asmocon A), Acuvue Oasys (senofilcon A) and Dailies Total 1 (delefilcon A). With the Dailies Total 1 material, it was presumed that the claimed transient water content also generates a gradient RI from the lens surface to the central bulk of the material, which makes it difficult to determine the critical angle. When attempting to measure the RI of the other three SiHy materials with a manual Abbe refractometer, a rather fuzzy dark-to-bright transition line was noticed, indicating that there are in-homogeneities on the lens surface. This could be due to some 'skin effect', as the material interacts with the polypropylene mould while curing, a surface coating or irregular polymerisation on the lens surface. For the Dailies Total 1 lens type, Schafer et al. [14] were able to measure the RI straight out from the blister pack using Metricon M-2010 Prism Coupler. Other studies [5,7] also experienced difficulties in measuring some particular lens materials which could be because of problem arising from a non-uniform RI at the lens surface.

In general, the RIs of the hydrogel lens materials were lower than those of the silicone hydrogel lens materials. This is most likely due to the fact that all the measured hydrogel materials had a higher water content than the silicone hydrogel materials. Extrapolating the water content correlation trends, low water content hydrogel materials would have similar RIs to corresponding silicone hydrogel materials.

The biggest difference between measured and the nominal RI as reported by the manufacturers, was found for Proclear 1-Day lens material of 0.014 and 0.0099 for measured in PBS and packaging solution, respectively. There were 5 PBS soaked and 3 packaging solution soaked lens materials which fell outside the ISO tolerance compared to their nominal RI. ISO 18369-4 allows RI measurements to be performed at wavelengths of either 546.1 or 587.6 nm. Manufacturers may choose to

specify their nominal RI values at wavelength of 587.6 nm, possibly explaining part of the difference with RI values measured at 546.1 nm in this study. However, very little is known or published on the dispersion properties of soft contact lens materials.

With most of the wet cell based power mapping instruments using the operating wavelength of 546 nm, measuring the RI at the same wavelength eliminates any dispersion induced power conversion errors, as discussed in the next paragraph. Other studies also measured and published RI measurement results for some of the soft contact lens materials using CLR 12-70 instrument. The biggest difference in RI between nominal and the measured reported by Gonzalez-Mejome et al. [7] was about 0.025 for hydrogel and 0.013 for silicone-hydrogel lens materials. Nichols et al. [5] found that the mean difference between nominal against the measured were not statistically different from zero. However, they have only measured a range of hydrogel lens materials but not included any silicone hydrogel materials. The lenses used in that study were soaked in 0.9 % sodium chloride saline before measuring their RIs. Schafer et al. [14] measured Biotrue ONEday straight out of the packaging solution to be (mean $\pm$ stdev) 1.375 ± 0.0008 between 10 lenses which is comparable to the 1.3739 ± 0.0003 measured in this study.

Comparing the measured RIs between the PBS and packaging solution soaked lens RI results, the PBS RIs were slightly higher with the means for 7 lens types being significantly different to packaging solution RIs. The RIs of all the packaging solution were measured, with Dailies Aqua Comfort Plus lens having the largest difference of 0.004 compared to the standard PBS of 1.334. Pearson [15] also measured the RI of the lens solutions using RFM870 digital refractometer for three readily available contact lens saline solutions. Although the measurements were made at wavelength 589 nm, the authors proposed corresponding RIs for wavelength of 546 nm used by focimeters in accordance with a study in which the refractive index of distilled water was measured at different wavelengths. The RI differences were up to 0.002 between the two wavelengths. In this study, the aim was to provide practitioners with information that closely relate to on-eye conditions, which is most accurately achieved using PBS as the standard measurement medium.

The back vertex power of soft contact lenses can be measured either in solution within a wet cell or in air. The major advantage of measuring the power in a wet cell is that the environment can be controlled. However, this means that the RI of the lens, and the solution the lens is soaked in, must be accurately known. Variations of the equation to

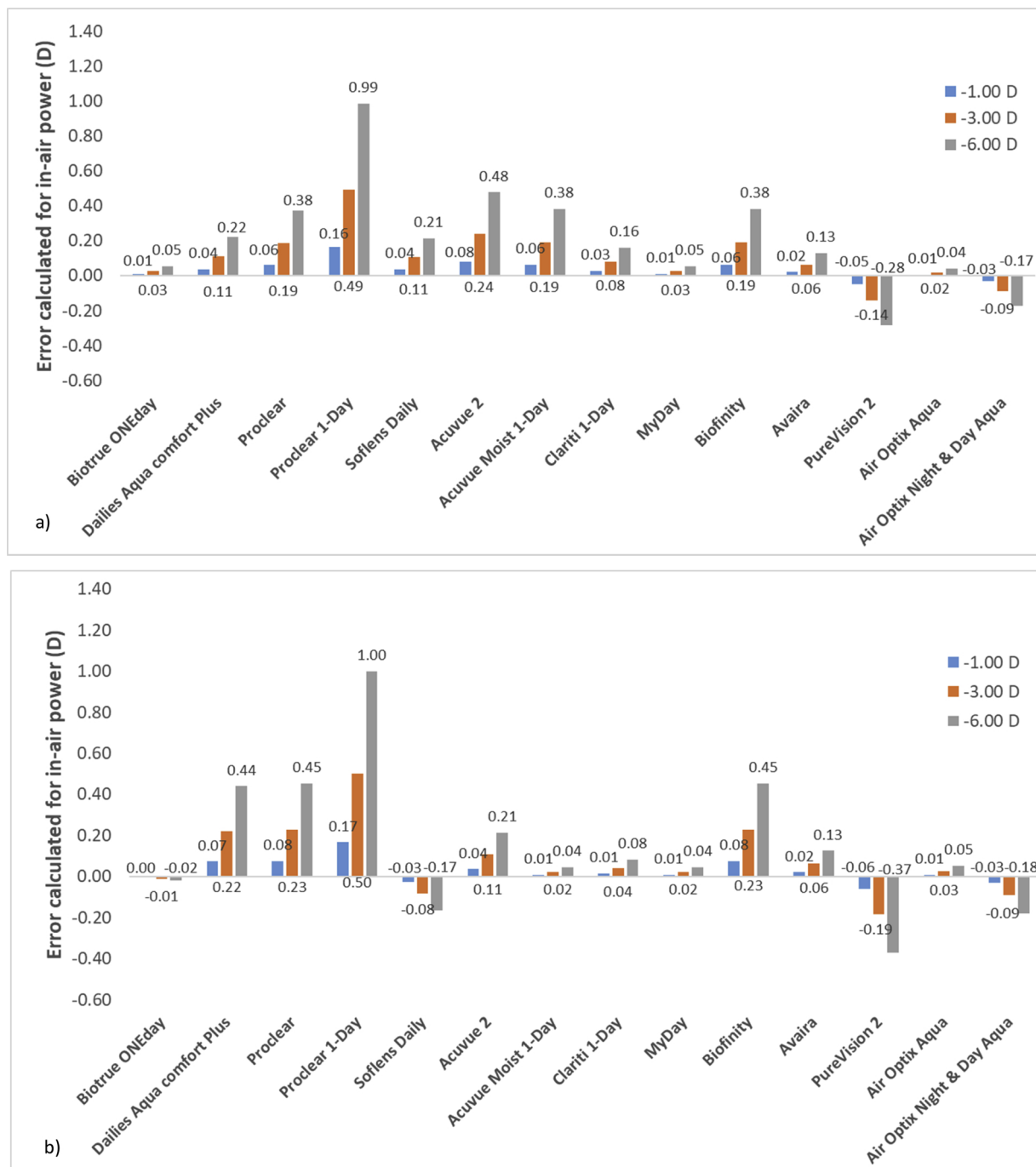


Fig. 3. Difference in-air power calculated for a -1.00 D, -3.00 D and -6.00 D lens when a) measured in PBS vs manufacturer's nominal RI and b) measured in packaging solution vs manufacturer's nominal RI.

convert the back vertex power measured in solution to its corresponding power in air have been previously published depending on the application and known variables. [9,11] Pearson [9] compared these equations extensively and investigated the errors as a consequence of inaccuracies in different variables used in the conversion of wet to in-air power calculations. The author concluded that measured power in saline can be reliably converted to power in air using the original equation of Wichterle [16] or those of Patel et al. [17] and Hough et al. [18]. Although, the inaccuracies in the RI of the lens and/or RI of the solution played greatest contributing factor when converting wet to in-air power. Campbell [11] reported that when applying the measured wet lens power to in-air power, an error of 0.015 in the RI can lead to an error of more than 1.00 D. In the current study, using the biggest

difference of RI between the PBS measured and nominal RI of the lens for Proclear 1-Day of 0.014, the error of the in-air power calculated would be 0.16 D, 0.49 D and 0.99 D for -1.00 D, -3.00 D and -6.00 D lens, respectively. As the nominal RIs reported by manufacturers are usually in two decimals only, they lack the accuracy for reliably converting optical powers, measured in a wet cell, to in-air power. Because of the ambiguities in the ISO standard 18369 with respect to the soaking solution and wavelength measurement conditions for soft contact lens back vertex power, the labelled power may not always be comparable between different lens types and manufacturers.

Gladstone and Dale [19] first described the linear relationship between RI and WC for hydrogel materials and the theory is now well established. However, for the silicone hydrogel materials, the same

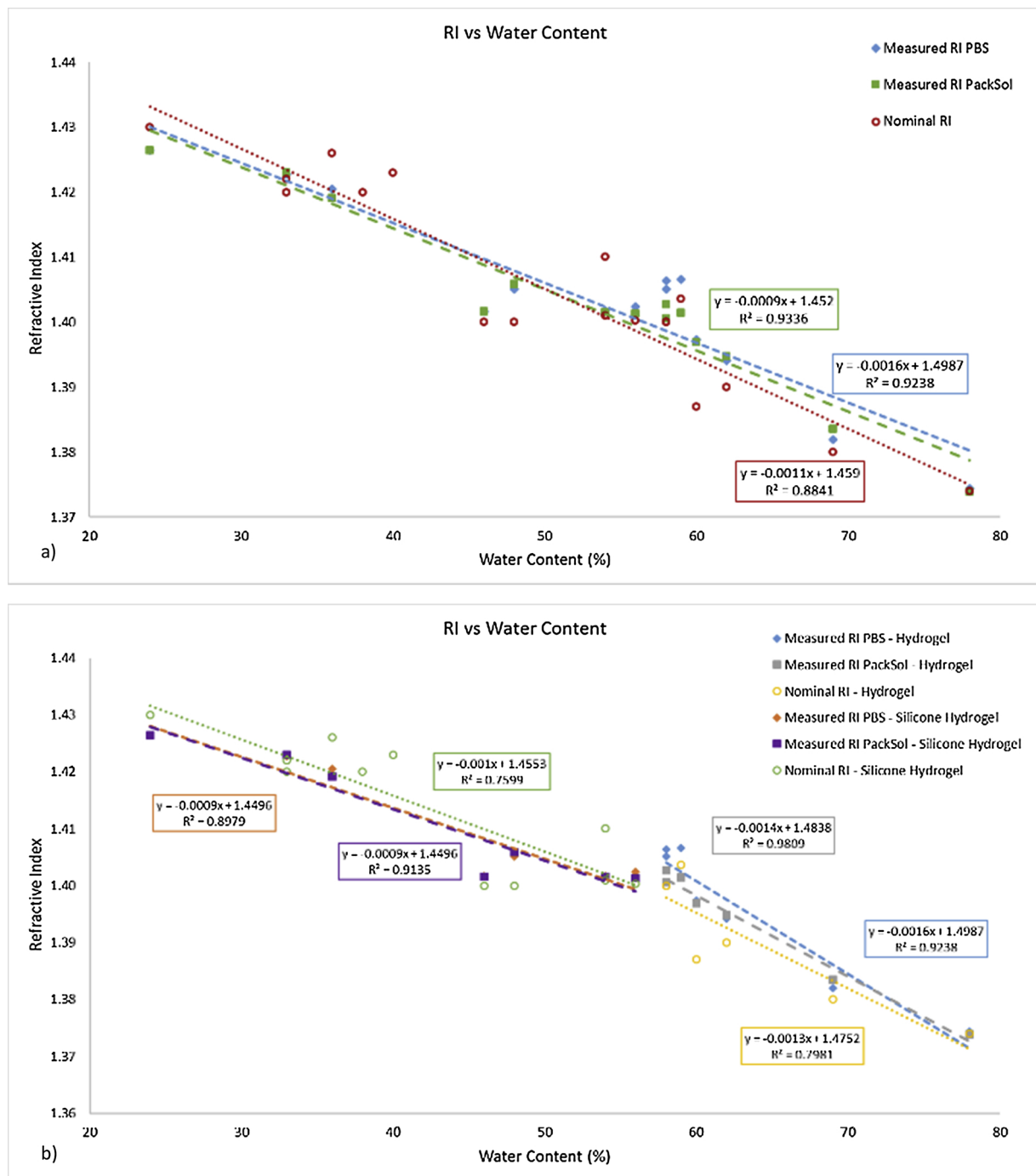


Fig. 4. Relationship between measured and nominal RI with the nominal WC for a) all lens types; b) separated hydrogel and silicone hydrogel lens types.

relationship is not well established. Gonzalez-Mejome et al. [7] plotted the relationships between the nominal RI and WC as well as the measured RI and WC for hydrogels and silicone hydrogels. The relationship between the nominal RI and WC did not correlate as well as the measured RI and nominal WC, which showed an excellent correlation of r -square value of 0.979. In the current study, the measured RI and nominal WC showed slightly higher correlation compared to the nominal RI and WC relationship. The same trend of decrease in RI as WC increases were observed for both hydrogels and silicone hydrogels.

One of the limitations of the Abbe refractometers is that they measure the refractive index on the lens surface with the assumption that the lens materials are homogenous with a uniform RI throughout the lens thickness and diameter. This may not necessarily be the case for all materials as small differences can be induced by the curing

process or due to a gradient WC as promoted by the recently introduced Dailies Total 1 lens. Another potential limitation of this, and most of the previous studies, is the limited selection of measured lens powers. A wider range of plus and minus lens powers need to be measured to confirm that there is no dependence of RI on lens power and correlated center thickness changes.

5. Conclusion

Though the CLR12-70 is generally a reliable and accurate digital refractometer for the measurement of soft contact lens materials, it failed to measure four of the silicone hydrogel lens type materials. Accurate RI measurements are of particular relevance with the increased use of wavefront sensors to measure lens power while soaked in

solution. Even small (< 0.001) errors in solution or material RI can lead to significant errors in converted in-air power. In this respect, manufacturer provided RI data with only two decimal places precision may not be sufficient. To obtain valid in-air lens power results, measurement conditions, including wavelength, must match the material and solution RIs used for the conversion.

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